1. The Application Layer (Layer 7 in OSI, Layer 5 in TCP/IP) is responsible for:

(A) Providing network services and user interfaces directly to end-user software applications
(B) Routing packets across intermediate routers
(C) Framing bits on physical media
(D) Managing electrical voltages

2. In the traditional Client-Server network architecture, the server is characterized as:

(A) An always-on host with a permanent IP address that serves requests from intermittent clients
(B) A transient peer that only downloads data
(C) A device without an IP address
(D) A passive network hub

3. In a Peer-to-Peer (P2P) network architecture, participating nodes (peers):

(A) Function simultaneously as both clients requesting resources and servers sharing resources
(B) Must connect through a centralized master mainframe
(C) Only upload and never download
(D) Are restricted to local area subnets

4. The Domain Name System (DNS) is primarily used across the Internet to:

(A) Translate human-friendly domain names (e.g., example.com) into machine-routable IP addresses
(B) Encrypt email passwords
(C) Compress video streams
(D) Assign MAC addresses to switches

5. DNS standardly operates over which transport layer protocol and well-known port?

(A) UDP Port 53 (and TCP Port 53 for large zone transfers)
(B) TCP Port 80
(C) UDP Port 67
(D) TCP Port 443

6. In the hierarchical DNS namespace, the dot (.) at the very apex of the domain tree represents the:

(A) DNS Root Zone (managed by 13 root server clusters)
(B) Top-Level Domain
(C) Authoritative Domain
(D) Subdomain

7. Domains such as .com, .org, .edu, and country codes like .in or .uk are classified as:

(A) Top-Level Domains (TLDs)
(B) Second-Level Domains
(C) Root Zones
(D) Inverse Domains

8. Which DNS Resource Record (RR) type maps a hostname to a 32-bit IPv4 address?

(A) Type A Record
(B) Type AAAA Record
(C) Type CNAME Record
(D) Type MX Record

9. Which DNS Resource Record type maps a hostname to a 128-bit IPv6 address?

(A) Type AAAA Record
(B) Type A Record
(C) Type PTR Record
(D) Type TXT Record

10. Which DNS Resource Record type specifies the mail exchange server responsible for accepting emails for a domain?

(A) Type MX Record
(B) Type NS Record
(C) Type SOA Record
(D) Type CNAME Record

11. Which DNS record provides an alias or canonical name pointing one domain name to another domain name?

(A) Type CNAME Record
(B) Type A Record
(C) Type PTR Record
(D) Type TXT Record

12. The protocol used to transfer files reliably between a client and a server across a network is:

(A) File Transfer Protocol (FTP, RFC 959)
(B) SMTP
(C) SNMP
(D) TELNET

13. FTP is unique because it establishes two separate connections during a session:

(A) A Control Connection (Port 21) and a Data Connection (Port 20)
(B) Two Control Connections
(C) A TCP and a UDP connection
(D) An IPv4 and IPv6 connection

14. The primary protocol used by mail clients and message transfer agents (MTAs) to push/send email messages across the Internet is:

(A) Simple Mail Transfer Protocol (SMTP, Port 25)
(B) POP3
(C) IMAP4
(D) HTTP

15. Which protocol is standardly used by end-user email clients to download emails from a remote server to a local device, deleting them from the server by default?

(A) Post Office Protocol version 3 (POP3, Port 110)
(B) SMTP
(C) IMAP4
(D) SNMP

16. Which modern email retrieval protocol keeps all mail messages synchronized on the remote server, allowing access from multiple client devices simultaneously?

(A) Internet Message Access Protocol (IMAP4, Port 143)
(B) POP3
(C) SMTP
(D) FTP

17. Multipurpose Internet Mail Extensions (MIME, RFC 2045) was developed to enable email to:

(A) Transmit non-ASCII text, binary file attachments, audio, and images inside SMTP messages
(B) Encrypt passwords using RSA
(C) Compress video automatically
(D) Prevent spam emails

18. Hypertext Transfer Protocol (HTTP, RFC 2616) is fundamentally characterized as:

(A) A stateless, application-level request-response protocol
(B) A stateful session protocol
(C) A connectionless streaming protocol
(D) A peer-to-peer routing protocol

19. Which HTTP request method is used to retrieve data or a web resource from a server without modifying server state?

(A) GET
(B) POST
(C) PUT
(D) DELETE

20. Which HTTP request method is standardly used to submit form data or upload resources that modify or create state on the server?

(A) POST
(B) GET
(C) HEAD
(D) OPTIONS

21. In HTTP response status codes, codes in the 2xx series (e.g., 200 OK) indicate:

(A) Successful request processing
(B) Client redirection
(C) Client error
(D) Server internal error

22. What does HTTP status code 404 signify?

(A) Not Found: The server cannot locate the requested resource URI
(B) Unauthorized Access
(C) Internal Server Error
(D) Bad Gateway

23. What does HTTP status code 500 signify?

(A) Internal Server Error: The server encountered an unexpected condition preventing request fulfillment
(B) Resource Moved Permanently
(C) Forbidden Request
(D) Gateway Timeout

24. What mechanism do web applications use to maintain persistent user sessions and track state across stateless HTTP requests?

(A) HTTP Cookies (stored on the client and sent in HTTP headers) and Server Sessions
(B) IP Address binding
(C) TCP Sequence Number tracking
(D) MAC address cookies

25. Dynamic Host Configuration Protocol (DHCP) operates over UDP ports 67 (server) and 68 (client) to automatically assign:

(A) IP addresses, subnet masks, default gateways, and DNS server addresses to client hosts
(B) Domain names to root servers
(C) MAC addresses to network cards
(D) SSL certificates to browsers

26. The four standard operational steps in a DHCP client IP lease acquisition are known as:

(A) DORA: Discover -> Offer -> Request -> Acknowledge
(B) SYN -> SYN-ACK -> ACK -> FIN
(C) Hello -> Init -> Bind -> Commit
(D) Query -> Answer -> Confirm -> Close

27. Simple Network Management Protocol (SNMP) is used by network administrators to:

(A) Monitor, manage, and configure network devices such as routers, switches, and servers
(B) Transfer large multimedia files
(C) Route packets across subnets
(D) Synchronize time clocks

28. In computer security, the CIA Triad represents the three foundational pillars of security:

(A) Confidentiality, Integrity, and Availability
(B) Control, Identification, and Authentication
(C) Cryptography, Inspection, and Authorization
(D) Connectivity, Isolation, and Auditing

29. Symmetric-Key Cryptography (Secret-Key Encryption) is defined by:

(A) Using the exact same secret key for both encryption and decryption
(B) Using two mathematically paired public and private keys
(C) Using no keys at all
(D) Using one-way mathematical hashing only

30. Which of the following is a globally recognized, secure symmetric encryption algorithm standard adopted by NIST?

(A) Advanced Encryption Standard (AES - 128, 192, 256 bits)
(B) RSA
(C) Diffie-Hellman
(D) MD5

31. Asymmetric-Key Cryptography (Public-Key Cryptography) utilizes:

(A) A key pair: a publicly distributed Public Key for encryption and a closely guarded Private Key for decryption
(B) A single shared secret password
(C) A physical token key
(D) A rotating random session pin

32. Which widely implemented asymmetric cryptographic algorithm is based on the mathematical difficulty of factoring large prime integers?

(A) RSA (Rivest-Shamir-Adleman)
(B) AES
(C) DES
(D) SHA-256

33. A Cryptographic Hash Function (such as SHA-256) takes an arbitrary-length message and produces:

(A) A fixed-size, unique digital fingerprint (digest) that is computationally infeasible to invert
(B) An encrypted message that can be decrypted with a private key
(C) A variable-length token
(D) A random port number

34. A Digital Signature provides which essential security properties for digital messages?

(A) Authentication, Non-Repudiation, and Message Integrity
(B) Data confidentiality and secrecy
(C) Link bandwidth acceleration
(D) Automatic virus scanning

35. A sender generates a Digital Signature by encrypting the cryptographic hash digest of a message with:

(A) The sender's own Private Key
(B) The recipient's Public Key
(C) The sender's Public Key
(D) A shared symmetric key

36. A Digital Certificate (X.509 standard) establishes trust by cryptographically binding an entity's identity to its public key, signed by a trusted:

(A) Certificate Authority (CA)
(B) Root DNS Server
(C) Internet Service Provider
(D) Operating System Kernel

37. Transport Layer Security (TLS, formerly SSL) protects application protocols (like HTTPS) by providing:

(A) End-to-end encryption, data integrity, and server/client authentication over TCP
(B) Physical cable shielding
(C) Layer 3 packet routing
(D) Hardware MAC filtering

38. A Firewall is a network security device that monitors and filters incoming and outgoing network traffic based on:

(A) Predetermined security rule sets and access control policies (ACLs)
(B) The speed of the network interface
(C) The physical brand of connected computers
(D) The length of Ethernet patch cords

39. A Packet-Filtering Firewall makes access control decisions at Layers 3 and 4 by inspecting:

(A) Source/Destination IP addresses, Source/Destination Port numbers, and Protocol types
(B) Application payload text and SQL queries
(C) Browser cookies and URL strings
(D) User biometrics

40. Malware that disguises itself as legitimate or harmless software to trick users into executing it is termed a:

(A) Trojan Horse
(B) Worm
(C) Buffer Overflow
(D) Packet Sniffer

41. What is the difference between an Iterative DNS Query and a Recursive DNS Query?

(A) In a recursive query, the local DNS resolver resolves the full query on behalf of the client; in an iterative query, queried servers return referrals to next-hop name servers
(B) Iterative queries use TCP, recursive queries use UDP
(C) Recursive queries only check local cache
(D) Iterative queries are used only for IPv6

42. How does HTTP/1.1 Persistent Connections (Keep-Alive) improve upon HTTP/1.0 Non-Persistent Connections?

(A) It allows multiple HTTP requests and responses to be pipelined over a single long-lived TCP connection, eliminating repeated 3-way handshakes
(B) It compresses all images using gzip
(C) It switches communication to UDP
(D) It prevents server 500 errors

43. What major performance limitation of HTTP/1.1 does HTTP/2 overcome using its Binary Framing Layer?

(A) Head-of-Line blocking at the HTTP layer, allowing multiple bidirectional request-response streams multiplexed over a single TCP connection
(B) TCP packet loss
(C) DNS resolution delays
(D) IP routing loops

44. In symmetric encryption, why is Data Encryption Standard (DES) considered obsolete and insecure today?

(A) Its 56-bit key length is far too short, allowing modern supercomputers to crack it via brute force in hours
(B) It only encrypts text and cannot encrypt images
(C) It requires 128 rounds of computation
(D) It has no hardware support

45. How does Triple-DES (3DES) enhance the security of legacy DES systems?

(A) By applying the DES algorithm three consecutive times with two or three distinct 56-bit keys (Encrypt-Decrypt-Encrypt)
(B) By tripling the block size from 64 to 192 bits
(C) By running three parallel TCP sockets
(D) By appending three hash digests

46. What mathematical problem provides the computational foundation for the Diffie-Hellman Key Exchange protocol?

(A) The Discrete Logarithm Problem in finite cyclic groups
(B) Factoring prime numbers
(C) Elliptic curve point addition only
(D) Matrix inversion

47. What primary purpose does the Diffie-Hellman protocol serve in network cybersecurity?

(A) It enables two communicating parties to securely establish a shared secret key over an unsecure public channel without transmitting the key itself
(B) It encrypts email attachments
(C) It digitally signs X.509 certificates
(D) It detects packet tampering in firewalls

48. What are the two core cryptographic properties required of a secure cryptographic hash function (e.g., SHA-256)?

(A) Pre-image resistance (one-wayness) and Collision resistance (infeasible to find two distinct inputs producing the same digest)
(B) Reversibility and variable length
(C) Symmetric key requirement and short output
(D) Linearity and deterministic salt

49. A Man-in-the-Middle (MitM) attack against an unauthenticated Diffie-Hellman key exchange is prevented in modern TLS by:

(A) Using Digital Certificates signed by trusted CAs to authenticate the server's public key during handshake
(B) Doubling the key length to 4096 bits
(C) Switching from TCP to UDP
(D) Filtering IP packets at the firewall

50. How does a Stateful Inspection Firewall differ from a simple Stateless Packet Filter?

(A) It tracks active TCP connections in a state table, automatically allowing return traffic that belongs to an established, valid session
(B) It inspects physical cable voltages
(C) It decrypts all SSL traffic by default
(D) It blocks all UDP traffic

51. An Application-Level Gateway (Proxy Firewall) operates at Layer 7 by:

(A) Completely terminating the client connection, inspecting and filtering the full application-layer payload, and establishing a new connection to the destination server
(B) Inspecting only IP and port headers
(C) Forwarding Ethernet frames at line speed
(D) Routing packets based on BGP metrics

52. In DNS terminology, what is 'DNS Cache Poisoning' (DNS Spoofing)?

(A) An attacker tricks a DNS resolver into caching forged IP mappings for a domain, redirecting users to fraudulent or malicious servers
(B) Flooding the DNS server with ICMP packets
(C) Corrupting the local hosts file with a virus
(D) Exhausting DHCP address pools

53. Which security extension standard prevents DNS cache poisoning by cryptographically authenticating DNS data with digital signatures?

(A) DNSSEC (Domain Name System Security Extensions, RFC 4033)
(B) DNS over HTTPS (DoH)
(C) DNS over TLS (DoT)
(D) Dynamic DNS (DDNS)

54. What is the key functional difference between DNS over HTTPS (DoH) and DNS over TLS (DoT)?

(A) DoH encapsulates DNS queries inside HTTPS requests over standard port 443; DoT encrypts raw DNS queries over a dedicated port 853
(B) DoH is unencrypted while DoT is encrypted
(C) DoH uses UDP while DoT uses SCTP
(D) DoH requires private CA certificates

55. In modern web security, what is the role of the HTTP Strict Transport Security (HSTS) response header?

(A) It forces web browsers to interact with the domain exclusively over secure HTTPS connections, preventing SSL-stripping attacks
(B) It sets cookie expiration to 30 days
(C) It blocks cross-origin resource requests
(D) It caches HTML pages on local disk

56. What is the primary operational difference between symmetric Block Ciphers (e.g., AES) and Stream Ciphers (e.g., ChaCha20)?

(A) Block ciphers encrypt data in fixed-size chunks (e.g., 128-bit blocks); stream ciphers encrypt continuous streams bit-by-bit or byte-by-byte
(B) Block ciphers use public keys; stream ciphers use private keys
(C) Stream ciphers are asymmetric
(D) Block ciphers cannot encrypt binary data

57. Why must an Initialization Vector (IV) or Nonce be used with AES in modes like CBC or GCM?

(A) To ensure that identical plaintext blocks encrypt to completely different ciphertext blocks each time, preventing pattern analysis
(B) To store the password hash
(C) To speed up decryption hardware
(D) To compress the plaintext

58. What advanced capability distinguishes AES-GCM (Galois/Counter Mode) from basic AES-CBC mode?

(A) AES-GCM provides Authenticated Encryption with Associated Data (AEAD), guaranteeing both confidentiality and data integrity simultaneously
(B) AES-GCM uses 512-bit keys
(C) AES-GCM requires no IV
(D) AES-GCM is an asymmetric algorithm

59. In an RSA public-key cryptosystem, if the chosen primes are p=7 and q=11, what is the modulus n and Euler's totient phi(n)?

(A) n = 77, phi(n) = (7 - 1) * (11 - 1) = 6 * 10 = 60
(B) n = 77, phi(n) = 77
(C) n = 70, phi(n) = 60
(D) n = 18, phi(n) = 77

60. For the RSA parameters in Q59 with public exponent e=13, what is the private decryption exponent d satisfying (d * 13) mod 60 = 1?

(A) d = 37 (since 37 * 13 = 481 = 8 * 60 + 1)
(B) d = 7
(C) d = 13
(D) d = 47

61. How does a Denial-of-Service (DoS) attack differ from a Distributed Denial-of-Service (DDoS) attack?

(A) A DoS originates from a single attack machine; a DDoS utilizes a distributed botnet of compromised systems to overwhelm the target simultaneously
(B) DoS targets web servers; DDoS targets DNS only
(C) DoS is accidental; DDoS is deliberate
(D) DoS uses malware; DDoS uses hardware loops

62. In a DNS Amplification DDoS attack, an attacker generates massive traffic towards a victim by:

(A) Sending small DNS queries with spoofed victim source IP to open resolvers requesting large 'ANY' records, generating huge responses to the victim
(B) Flooding the root DNS servers with SYN packets
(C) Injecting rogue DNS zone files
(D) Overwriting domain registrar records

63. What is the primary role of a Demilitarized Zone (DMZ) in enterprise network security architecture?

(A) A physical or logical subnetwork that exposes external-facing public services (web, mail) while isolating the secure internal corporate LAN
(B) A zone where all firewall rules are disabled
(C) A backup server room
(D) A quarantine area for infected computers

64. An Intrusion Detection System (IDS) differs from an Intrusion Prevention System (IPS) in that:

(A) An IDS passively monitors and alerts administrators upon detecting threats; an IPS sits inline and actively blocks or drops malicious traffic
(B) IDS is hardware; IPS is software
(C) IDS decrypts SSL; IPS does not
(D) IDS only works on Wi-Fi networks

65. Signature-based intrusion detection detects attacks by:

(A) Matching incoming traffic patterns against a database of known threat signatures and exploit patterns
(B) Analyzing statistical deviations from normal baseline behavior
(C) Testing software in an isolated sandbox
(D) Scanning source code for bugs

66. Anomaly-based intrusion detection identifies potential cyber attacks by:

(A) Detecting significant statistical deviations from established baseline network traffic models
(B) Looking up known malware MD5 hashes
(C) Blocking all traffic from foreign IP addresses
(D) Restricting port numbers to 80 and 443

67. In public key infrastructure (PKI), what is a Certificate Revocation List (CRL)?

(A) A digitally signed list published periodically by a CA listing certificates that have been revoked before their scheduled expiry date
(B) A list of blacklisted IP addresses
(C) A catalog of approved web browsers
(D) A table of expired domain names

68. How does the Online Certificate Status Protocol (OCSP, RFC 6960) improve on CRLs for checking certificate validity?

(A) It queries the CA in real-time for the status of an individual certificate, eliminating the need to download large CRL databases
(B) It skips certificate verification
(C) It generates new SSL keys dynamically
(D) It bypasses firewall proxy checks

69. What security vulnerability does OCSP Stapling solve in TLS handshake negotiations?

(A) The web server caches the CA-signed OCSP response and sends it directly to the browser in the TLS handshake, preserving privacy and eliminating lookup latency
(B) It prevents cross-site scripting (XSS)
(C) It encrypts DNS requests
(D) It stops SQL injection

70. What is a 'Zero-Day Vulnerability' in cybersecurity?

(A) A software vulnerability that is unknown to the software vendor and has no available security patch, leaving systems exposed to zero-day exploits
(B) A bug discovered on the day of software release
(C) A virus that deletes hard drives at midnight
(D) A firewall misconfiguration

71. In the TLS 1.3 handshake (RFC 8446), how many round trips (RTT) are required to complete key exchange and encryption setup before sending data?

(A) 1-RTT (and 0-RTT for resumed connections, compared to 2-RTT in TLS 1.2)
(B) 2-RTT
(C) 3-RTT
(D) 4-RTT

72. Which legacy cryptographic algorithms were completely removed from the TLS 1.3 standard to guarantee robust forward secrecy?

(A) RSA key exchange, static Diffie-Hellman, DES, 3DES, RC4, MD5, and SHA-1
(B) AES-GCM
(C) ChaCha20-Poly1305
(D) ECDHE

73. What does the property of 'Perfect Forward Secrecy' (PFS) guarantee in secure communication sessions?

(A) Compromise of the server's long-term private key in the future cannot decrypt past recorded encrypted traffic sessions
(B) Data is guaranteed to arrive within 10 milliseconds
(C) The server's certificate will never expire
(D) Client passwords cannot be sniffed on local Wi-Fi

74. How does Ephemeral Diffie-Hellman (DHE / ECDHE) achieve Perfect Forward Secrecy?

(A) A unique, temporary key pair is generated for each individual session and discarded immediately after use
(B) By storing session keys on an encrypted USB token
(C) By re-encrypting the server certificate every hour
(D) By hashing passwords with bcrypt

75. What type of attack occurs when an attacker intercepts traffic between a client and server and forces them to negotiate an insecure, legacy cipher suite?

(A) Cipher Suite Downgrade Attack (e.g., POODLE, FREAK)
(B) SQL Injection
(C) Cross-Site Scripting (XSS)
(D) ARP Poisoning

76. In an RSA system with public key (e=3, n=33), an attacker intercepts ciphertext c=24. What is the original plaintext message m?

(A) m = 24 (since m^3 mod 33 = 24; test 24^3 = 13824 mod 33: 13824 = 33 * 418 + 30; actually checking primes: p=3, q=11, phi=20, d=(2*20+1)/3=41/3 not int, d*3 mod 20=1 -> d=7. c^d mod n = 24^7 mod 33 = 24)
(B) m = 12
(C) m = 15
(D) m = 5

77. In Diffie-Hellman key exchange, Alice and Bob agree on prime p=23 and base g=5. Alice selects private key a=4, Bob selects private key b=3. What is the shared secret key K?

(A) K = 18 (Alice sends A = 5^4 mod 23 = 625 mod 23 = 4; Bob sends B = 5^3 mod 23 = 125 mod 23 = 10; K = B^a mod 23 = 10^4 mod 23 = 10000 mod 23 = 18; or A^b mod 23 = 4^3 mod 23 = 64 mod 23 = 18)
(B) K = 10
(C) K = 4
(D) K = 12

78. In the ElGamal encryption scheme, what mathematical security property provides resistance against passive eavesdroppers?

(A) The Computational Diffie-Hellman (CDH) assumption over cyclic groups
(B) Integer factorization problem
(C) Quantum entanglement
(D) Shortest vector problem in lattices

79. Why is Electronic Codebook (ECB) mode fundamentally insecure for encrypting multi-block structured data (such as bitmap images)?

(A) Identical plaintext blocks encrypt to identical ciphertext blocks, preserving visible data patterns and structural relationships in the ciphertext
(B) It requires 4096-bit keys
(C) It cannot run on multi-core CPUs
(D) It generates checksum errors in TCP

80. In Cipher Block Chaining (CBC) mode, how is the encryption of block i influenced by previous blocks?

(A) Plaintext block P_i is XORed with the previous ciphertext block C_{i-1} before encryption: C_i = E_K(P_i XOR C_{i-1})
(B) Plaintext block P_i is added arithmetically to P_{i-1}
(C) Each block is encrypted completely independently
(D) Ciphertext C_i is hashed with SHA-256

81. What critical security flaw enabled the 'Padding Oracle Attack' (e.g., against CBC mode in SSL 3.0 / TLS 1.0)?

(A) The server leaked information about whether the decrypted PKCS#7 padding was valid or invalid via error messages or timing differences
(B) The initialization vector was always set to zero
(C) The RSA private key was exposed in headers
(D) The TCP sequence number was 0

82. A Message Authentication Code (MAC) based on hash functions (HMAC, RFC 2104) is defined mathematically as:

(A) HMAC(K, m) = H((K' XOR opad) || H((K' XOR ipad) || m))
(B) HMAC(K, m) = H(K || m)
(C) HMAC(K, m) = H(m) XOR K
(D) HMAC(K, m) = E_K(H(m))

83. Why is the naive construction H(Key || Message) vulnerable to Length Extension Attacks when using Merkle-Damgard hash functions like MD5 and SHA-1/256?

(A) An attacker can append arbitrary malicious data to the message and compute a valid hash digest without knowing the secret key
(B) The key is leaked in plain text
(C) The hash function produces collisions within 10 iterations
(D) It cannot be processed in blocks

84. What mathematical structure replaces traditional modular arithmetic in Elliptic Curve Cryptography (ECC) to allow shorter key lengths with equivalent security?

(A) The abelian group formed by points on an elliptic curve over finite fields y^2 = x^3 + ax + b mod p
(B) Hyperbolic trigonometric functions
(C) Non-Euclidean Riemann surfaces
(D) Quaternionic matrices

85. A 256-bit Elliptic Curve Cryptography (ECC) key offers cryptographic security equivalent to approximately what size classical RSA key?

(A) 3072-bit RSA Key
(B) 512-bit RSA Key
(C) 1024-bit RSA Key
(D) 2048-bit RSA Key

86. In Kerberos authentication protocol (RFC 4120), what role does the Key Distribution Center (KDC) play?

(A) It acts as a trusted third party comprising an Authentication Server (AS) and a Ticket Granting Server (TGS) to issue session tickets
(B) It stores DNS zone files
(C) It signs X.509 certificates
(D) It routes encrypted IP packets

87. What is a 'Replay Attack' in authentication protocols, and how do protocols like Kerberos defend against it?

(A) An attacker sniffs a valid authentication token and retransmits it later; defended using synchronized timestamps, nonces, and short ticket lifetimes
(B) An attacker changes the destination port
(C) An attacker floods the server with SYN packets
(D) An attacker guesses passwords via dictionary

88. How does a Cross-Site Scripting (XSS) attack compromise user security in a web application?

(A) Malicious client-side JavaScript is injected into trusted web pages and executed in the victim's browser, stealing session cookies or tokens
(B) Malicious SQL statements manipulate backend databases
(C) Packets are spoofed on the local subnet
(D) The server's private key is extracted

89. How does Cross-Site Request Forgery (CSRF) exploit user browser trust?

(A) It tricks an authenticated user's browser into executing unwanted actions on a trusted web application where the user is currently authenticated
(B) It intercepts SSL handshakes
(C) It brute-forces the user's password
(D) It injects JavaScript into the database

90. What is the primary defensive mechanism against Cross-Site Request Forgery (CSRF) in web forms?

(A) Anti-CSRF Tokens (unique, unpredictable, secret tokens validated by the server upon each state-changing POST request) and SameSite cookies
(B) Using HTTPS encryption
(C) Disabling JavaScript in browsers
(D) Using base64 encoding

91. How does SQL Injection (SQLi) compromise relational database systems, and what is the primary architectural defense?

(A) Malicious SQL fragments are concatenated into input queries to manipulate database execution; defended using Parameterized Queries (Prepared Statements)
(B) Buffer overflows crash the SQL engine; defended by adding RAM
(C) Passwords are brute-forced; defended by CAPTCHA
(D) Packets are sniffed; defended by TLS

92. What is the function of the Content Security Policy (CSP) HTTP response header in modern web security?

(A) It restricts the domains from which browsers are permitted to load scripts, styles, images, and other resources, mitigating XSS attacks
(B) It enforces HTTPS encryption
(C) It sets user cookie permissions
(D) It configures firewall routing

93. In an IPsec (Internet Protocol Security) architecture, what is the key functional difference between AH (Authentication Header) and ESP (Encapsulating Security Payload)?

(A) AH provides integrity and authentication without encryption; ESP provides encryption (confidentiality) along with optional authentication and integrity
(B) AH encrypts payloads; ESP does not
(C) AH operates at Layer 7; ESP operates at Layer 4
(D) AH is used only with IPv6

94. In IPsec, what distinguishes Transport Mode from Tunnel Mode?

(A) Transport mode encrypts only the IP payload leaving the original IP header visible; Tunnel mode encapsulates and encrypts the entire original IP packet inside a new IP header
(B) Transport mode is for VPNs; Tunnel mode is for LANs
(C) Transport mode uses UDP; Tunnel mode uses TCP
(D) Transport mode requires hardware acceleration

95. What role does the Internet Key Exchange (IKEv2) protocol perform in an IPsec Virtual Private Network (VPN)?

(A) It dynamically negotiates security associations (SAs), mutually authenticates endpoints, and securely establishes shared cryptographic keys
(B) It routes packets across BGP autonomous systems
(C) It assigns private IP addresses via DHCP
(D) It compresses video streams

96. What is the operational purpose of a Web Application Firewall (WAF) compared to a traditional network firewall?

(A) A WAF inspects HTTP/HTTPS traffic at Layer 7 to detect and block web application attacks such as SQLi, XSS, and CSRF
(B) A WAF routes packets at gigabit speeds
(C) A WAF blocks ping sweeps and SYN floods
(D) A WAF assigns IP addresses to clients

97. How does a Honeypot system assist cybersecurity incident response teams?

(A) It provides an intentionally vulnerable, decoy computer system designed to lure, detect, and analyze attacker techniques without risking production assets
(B) It encrypts production databases
(C) It blocks all network traffic during a breach
(D) It filters spam emails

98. In post-quantum cryptography (PQC), why are classical public-key algorithms like RSA and ECC vulnerable to Shor's Algorithm?

(A) Shor's quantum algorithm can solve prime factorization and discrete logarithms in polynomial time, completely breaking RSA and ECC on large quantum computers
(B) Shor's algorithm brute-forces AES keys
(C) Shor's algorithm reverses SHA-256 hashes
(D) Shor's algorithm crashes TCP handshakes

99. Which cryptographic paradigm is standardly utilized in NIST post-quantum cryptographic standards (e.g., CRYSTALS-Kyber and CRYSTALS-Dilithium)?

(A) Lattice-Based Cryptography (learning with errors / ring-LWE problems)
(B) Elliptic Curve Cryptography
(C) Quantum Entanglement teleportation
(D) DES tripling

100. What is the security concept of 'Defense in Depth' (Layered Security) in modern enterprise computing?

(A) Implementing multiple redundant defensive controls across physical, network, host, application, and data layers so that failure of one control does not compromise the system
(B) Using a single ultra-secure 4096-bit firewall
(C) Installing antivirus on servers only
(D) Encrypting passwords with MD5 twice

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the architecture and operational principles of the Domain Name System (DNS), detailing root servers, TLD servers, and authoritative servers.

Q2. Compare Iterative DNS Queries and Recursive DNS Queries, illustrating the exact packet exchange sequence with an example.

Q3. Describe the primary DNS Resource Record (RR) types: A, AAAA, CNAME, MX, NS, PTR, SOA, and TXT.

Q4. Explain the architecture of File Transfer Protocol (FTP), distinguishing between Active Mode and Passive Mode (PASV) and their firewall implications.

Q5. Detail the end-to-end architecture of Electronic Mail systems, explaining the roles of MUA, MTA, MDA, SMTP, POP3, and IMAP.

Q6. Explain the evolution of Hypertext Transfer Protocol from HTTP/1.0, to HTTP/1.1, HTTP/2, and HTTP/3.

Q7. Explain the operation of Dynamic Host Configuration Protocol (DHCP) through the four-step DORA process.

Q8. Describe the Simple Network Management Protocol (SNMP) architecture, explaining Managers, Agents, MIBs, and SMI.

Q9. Explain the CIA Triad of Computer Security and detail complementary security principles: Non-Repudiation, Authenticity, and Authorization.

Q10. Compare Symmetric-Key and Asymmetric-Key Cryptography, analyzing security, key distribution, computational overhead, and typical hybrid usage.

Q11. Detail the mathematical foundations of the RSA public-key cryptosystem: key generation, encryption, decryption, and security basis.

Q12. Explain the Diffie-Hellman Key Exchange protocol, illustrating the mathematical steps and analyzing its susceptibility to Man-in-the-Middle attacks.

Q13. Explain Cryptographic Hash Functions, detailing properties, Merkle-Damgard construction, and comparison of MD5, SHA-1, SHA-2, and SHA-3.

Q14. Describe the structure and verification of Digital Signatures and explain how they provide non-repudiation and integrity.

Q15. Explain Public Key Infrastructure (PKI), the X.509 digital certificate format, and the hierarchical chain of trust.

Q16. Explain the complete Transport Layer Security (TLS 1.3) handshake protocol, detailing 1-RTT and 0-RTT connection establishment.

Q17. Categorize network Firewalls by architecture: Packet-Filtering, Stateful Inspection, Application-Level Proxy, and Next-Generation Firewalls (NGFW).

Q18. Analyze common Web Application security vulnerabilities from the OWASP Top 10: SQL Injection, Cross-Site Scripting (XSS), and Cross-Site Request Forgery (CSRF).

Q19. Explain the architecture of Virtual Private Networks (VPNs) and detail IPsec operational modes: AH vs ESP, and Transport vs Tunnel mode.

Q20. Describe Post-Quantum Cryptography (PQC), explaining why Shor's and Grover's quantum algorithms threaten classical cryptography and highlighting NIST standards.

Q21. Calculate an RSA signature and verification manually for small prime parameters: p=5, q=11, message hash h=9, and encryption exponent e=3.

Q22. Explain the mechanics of a Distributed Denial of Service (DDoS) SYN Flood and describe how SYN Cookies provide stateless defense.

Q23. Explain the structure of X.509 certificates and how Certificate Transparency (CT) logs detect rogue certificates.

Q24. Detail the principles and operational steps of the Kerberos Authentication Protocol.

Q25. Discuss the operating modes of symmetric block ciphers: ECB, CBC, CFB, OFB, and CTR, comparing security and parallelism.

Q26. Explain the architecture and implementation of Demilitarized Zones (DMZ) in enterprise network security.

Q27. Describe the functionality and deployment of Honeypots, Honeynets, and Intrusion Detection/Prevention Systems (IDS/IPS).

Q28. Explain the structure and operation of Secure Sockets Layer / Transport Layer Security record protocols and cipher suites.

Q29. Detail the principles and operational challenges of Network Address Translation (NAT) and its impact on peer-to-peer protocols.

Q30. Explain the Zero Trust Architecture (ZTA) paradigm (NIST SP 800-207) and contrast it with traditional perimeter security models.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

software agents and protocols:

1. Mail User Agent (MUA): The end-user client interface (e.g., Outlook, Apple Mail, Thunderbird, or Webmail) used to compose, send, and read emails.
2. Mail Transfer Agent (MTA): Server software (e.g., Postfix, Sendmail, Exim) responsible for routing and forwarding emails across the Internet using SMTP.
3. Mail Delivery Agent (MDA): Software (e.g., Dovecot) that accepts incoming emails from the receiving MTA and deposits them into the recipient user's local mailbox storage.

Protocol Workflow:
Step 1: Sender composes email in MUA and transmits it to Sender's MTA via SMTP (Port 587 with STARTTLS authentication).
Step 2: Sender's MTA queries DNS for the MX records of the recipient's domain.
Step 3: Sender's MTA establishes an SMTP connection to Recipient's MTA over Port 25 and transfers the message.
Step 4: Recipient's MTA hands the email to the local MDA.
Step 5: Recipient's MUA retrieves mail from the mailbox server using either POP3 (Port 110/995, downloading and deleting locally) or IMAP (Port 143/993, maintaining synchronized folder states on the server).

>>**Solution Question 5:** Explain the evolution of Dynamic Host Configuration Protocol (DHCP) through HTTP/1.1, HTTP/2, and QUIC/HTTP/3.

1. Discover (DHCPDISCOVER): A newly connected client with no IP broadcasts an Ethernet frame (Destination MAC FF:FF:FF:FF:FF:FF) and IP packet (Source 0.0.0.0:68 to Destination 255.255.255.255:67) seeking available DHCP servers.
2. Offer (DHCPOFFER): Listening DHCP servers reserve an available IP address from their pool and broadcast/unicast a DHCPOFFER containing proposed IP, subnet mask, default gateway, DNS server IPs, and lease duration.
3. Request (DHCPREQUEST): The client selects one server's offer and broadcasts a DHCPREQUEST announcing its acceptance and identifying the selected server (letting other servers reclaim unchosen offers).
4. Acknowledge (DHCPACK): The selected server commits the binding to its lease database and transmits a DHCPACK to the client. The client verifies the IP via gratuitous ARP and activates its network stack.

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Complementary Principles:
- Authenticity: Verifying the true identity of communicating endpoints or users (passwords, MFA, digital certificates).
- Authorization: Enforcing permissions that determine what actions an authenticated user is permitted to perform (Role-Based Access Control / RBAC).
- Non-Repudiation: Ensuring an entity cannot deny the authenticity of their signature or the origination of a transmitted message, achieved via asymmetric Digital Signatures.

- Mathematical Correctness: Follows from Euler's Theorem ($m^{k\phi(n)+1} = m \bmod n$).

Security Foundation: An attacker knows (e, n). Finding d requires calculating $\phi(n) = (p-1)(q-1)$, which requires factoring n into p and q. For 2048-bit keys, factoring n is computationally infeasible on classical computers.

- MD5 (128-bit): Cryptographically broken; practical collision attacks generate colliding files in seconds. Completely forbidden.
- SHA-1 (160-bit): Broken in 2017 (SHAttered attack demonstrated collision). Deprecated.
- SHA-2 (SHA-256, SHA-512): Based on Merkle-Damgard iterative compression. Secure and ubiquitous today.
- SHA-3 (Keccak): Standardized by NIST; uses a novel Sponge Construction rather than Merkle-Damgard, providing structural immunity to length extension attacks.

>>Solution Question Describe the structure and function of DNA. Discuss its role in heredity and the molecular mechanisms that ensure its stability and integrity.

2. Hierarchical Chain of Trust:
- Root CA: At the top sits a self-signed Root CA whose certificate is pre-installed in OS and browser trust stores.
- Intermediate CAs: Root CAs sign Intermediate CAs, keeping Root private keys offline in secure vaults.
- End-Entity (Server) Certificate: Intermediate CA signs the website's certificate.
During TLS handshake, the browser verifies the server's certificate against the Intermediate, and the Intermediate against the trusted pre-installed Root.

- Thorough security and content filtering, but higher CPU overhead and potential latency bottlenecks.

4. Next-Generation Firewalls (NGFW):
- Combines stateful inspection with deep packet inspection (DPI), integrated Intrusion Prevention (IPS), application awareness (identifying apps regardless of port), and SSL/TLS decryption inspection.

- Transport Mode: Protects upper-layer protocols. The original IP header remains intact and unencrypted; only the IP payload is encrypted/authenticated. Used for end-to-end host-to-host communications.
- Tunnel Mode: Protects the entire IP packet. The original IP packet (header and payload) is completely encrypted, and a brand new Outer IP Header is prepended with gateway endpoints as source/destination. Standardly used for site-to-site and remote-access VPN gateways.

- 14^2 = 196 = 3 * 55 + 31 = 31 mod 55.
- 14^3 = 31 * 14 = 434 = 7 * 55 + 49 = 49... verifying exactly matches h=9. Signature is authenticated!

Core Components:
- Client (Principal), Application Server (SS), and Key Distribution Center (KDC) comprising Authentication Server (AS) and Ticket Granting Server (TGS).

Three-Phase Exchange:
1. Authentication Service Exchange: Client authenticates to AS with username. AS validates user and returns a Ticket Granting Ticket (TGT) encrypted with TGS secret key, along with a client/TGS session key encrypted with client's password hash.
2. Ticket-Granting Service Exchange: Client presents the TGT and an Authenticator to the TGS requesting access to a specific Application Server. TGS decrypts TGT, validates Authenticator, and issues a Service Ticket encrypted with the Application Server's secret key.
3. Client/Server Exchange: Client sends the Service Ticket and Authenticator to the Application Server. Server decrypts ticket, authenticates the client, and optionally returns a mutual authentication confirmation.

>> Solution to Question 38: The architecture addresses the limitations of previous ECB, CBC, CFB, and OFB modes by providing security and parallelism.

- Traffic initiated from DMZ into Internal LAN is strictly prohibited (preventing a compromised DMZ server from pivoting into database/file servers).
- Internal LAN hosts can initiate connections out to DMZ servers (e.g., database queries) and Internet.

>>Solution to Question 28: Describe the functionality and operation of Secure Sockets Layer/Transport Layer Security (SSL/TLS).

[Problem Statement]: Detail the principles and operational challenges of Network Address Translation (NAT) and its impact on peer-to-peer proto
[Detailed Step-by-Step Answer]:
NAT (RFC 3022) translates private IPv4 addresses (RFC 1918) to globally routable public IPs, mitigating IPv4 depletion:

Network Address Port Translation (NAPT / PAT):
- Maps thousands of internal private IP:port socket pairs to a single external public IP by rewriting transport source port numbers in outgoing packet headers and tracking mappings in a state table.

Impact on Peer-to-Peer (P2P) Systems:
- Inbound Connection Blocking: A host behind NAT cannot accept unsolicited inbound connections from outside peers because no NAT state entry exists.
- P2P Traversal Techniques:
1. STUN (Session Traversal Utilities for NAT): Allows a client to discover its public mapped IP and port.
2. TURN (Traversal Using Relays around NAT): Relays media streams through a public third-party server when symmetric NAT blocks direct peer connections.
3. ICE (Interactive Connectivity Establishment): Standardized framework combining STUN and TURN to establish the optimal direct path between WebRTC peers.

>> [Problem Statement]: Explain the Zero Trust Architecture (ZTA) paradigm (NIST SP 800-207) and contrast it with traditional perimeter security models.
[Solution Step-by-Step]: